

LONGVILLE, MN, USA

WMO: 727497

Lat: 46.990N

Lon: 94.204W

Elev: 1334

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04984

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-17.4	-29.1	1.1	-24.0	-24.1	1.5	-15.5	18.0	16.3	16.4	15.3	1.5	300	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	84.3	71.0	81.9	69.2	80.6	68.0	74.4	82.0	72.1	79.2	70.1	76.7	8.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	125.1	79.7	70.1	116.3	77.4	67.6	106.7	74.9	38.7	82.0	36.8	78.3	34.9	77.1	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.9	12.5	-30.7	89.8	6.8	2.2	-35.6	91.4	-39.5	92.7	-43.3	93.9	-48.2	95.5		
			-30.7	77.6	6.5	2.2	-35.5	79.2	-39.3	80.5	-43.0	81.7	-47.7	83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	9.2	12.1	26.1	40.2	52.4	61.8	66.9	64.3	56.9	43.4	29.7	15.4
	DBStd	22.58	14.53	13.14	13.40	9.66	8.49	6.26	5.57	5.96	8.34	8.49	10.73	12.79
	HDD50	5412	1265	1061	745	314	70	2	0	1	27	243	611	1073
	HDD65	9356	1730	1481	1207	743	401	131	44	86	266	670	1059	1538
	CDD50	1770	0	0	3	21	145	357	525	443	235	39	2	0
	CDD65	240	0	0	0	0	11	36	104	64	24	1	0	0
	CDH74	2444	0	0	0	9	187	409	1000	602	217	20	0	0
	CDH80	494	0	0	0	0	41	71	218	121	40	3	0	0
	WSAvg	4.9	5.0	4.9	5.1	5.4	5.2	4.4	4.0	3.8	4.8	5.5	5.5	4.8
	PrecAvg	28.0	0.7	0.6	1.3	2.1	3.2	4.3	4.1	3.1	3.1	3.1	1.3	0.9
Precipitation	PrecMax	33.7	1.9	1.9	3.5	5.2	7.6	7.5	7.6	6.3	7.1	6.0	4.1	2.1
	PrecMin	20.8	0.1	0.0	0.4	0.8	1.2	1.5	1.9	0.9	0.7	0.4	0.2	0.3
	PrecStd	3.3	0.5	0.4	0.9	1.0	1.5	1.6	1.6	1.5	1.3	1.6	0.9	0.5
	PrecStd	3.3	0.5	0.4	0.9	1.0	1.5	1.6	1.6	1.5	1.3	1.6	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	46.3	64.1	73.3	84.3	85.7	89.9	88.1	84.3	75.5	63.2	43.4
		MCWB	38.3	39.1	53.1	56.7	65.0	72.3	76.2	73.3	69.1	60.1	52.0	37.7
	2%	DB	36.5	39.4	55.3	67.7	79.4	81.7	84.4	82.3	80.6	69.6	55.2	38.6
		MCWB	33.2	34.9	46.8	52.7	60.8	68.5	72.2	70.6	68.6	57.2	48.2	35.7
	5%	DB	32.5	36.1	48.4	63.4	73.3	79.2	82.0	80.9	75.3	63.3	50.1	35.6
		MCWB	31.0	32.8	42.0	48.5	57.6	65.5	70.2	69.0	65.5	54.0	44.7	33.4
	10%	DB	28.4	32.0	44.6	57.5	70.0	75.2	80.6	77.2	72.3	57.1	45.3	32.2
		MCWB	27.1	30.1	38.9	45.7	56.4	63.5	69.1	66.6	64.0	49.5	40.7	31.2
	0.4%	WB	38.2	40.5	57.4	58.7	69.2	75.4	77.7	75.5	73.6	62.9	54.9	39.3
		MCDB	43.9	46.7	61.7	71.6	80.0	83.0	85.2	83.6	80.1	69.2	58.8	41.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.0	35.8	47.8	53.6	64.5	71.4	75.1	73.1	70.3	59.0	49.2	35.9
		MCDB	36.5	39.2	53.8	63.4	72.6	78.6	82.9	80.3	76.8	67.5	55.8	38.5
	5%	WB	30.7	32.7	42.9	50.2	61.5	68.6	72.7	71.0	67.5	54.8	44.8	33.3
		MCDB	32.3	35.7	48.7	61.0	69.2	75.6	80.0	77.8	74.3	60.6	49.0	34.6
	10%	WB	27.8	29.9	38.8	46.9	58.5	66.0	70.4	68.8	65.2	51.5	40.7	30.8
		MCDB	28.4	31.7	43.4	56.7	67.2	72.4	76.8	75.1	70.8	56.5	45.3	32.0
Mean Daily Temperature Range	5% DB	MDBR	19.5	21.6	21.5	22.1	23.2	21.4	21.5	21.8	21.6	17.7	15.8	15.8
		MCDBR	19.4	22.2	26.2	31.6	32.0	25.5	24.2	24.7	25.5	27.6	24.3	17.5
		MCWBR	16.1	17.2	17.1	18.0	15.9	13.6	13.3	13.5	15.1	16.1	16.1	14.3
	5% WB	MCDBR	18.3	20.1	24.2	28.5	25.3	21.6	21.5	21.4	22.0	24.4	22.6	15.6
		MCWBR	15.5	15.7	16.6	17.3	14.6	13.3	13.3	12.7	13.8	15.2	16.3	13.5
Clear-Sky Solar Irradiance	taub		0.255	0.280	0.307	0.352	0.370	0.372	0.379	0.381	0.352	0.309	0.282	0.263
	taud		2.369	2.302	2.388	2.360	2.367	2.389	2.387	2.399	2.459	2.572	2.518	2.394
	Ebn at Noon		270	283	292	286	284	283	279	274	273	272	259	252
	Edh at Noon		22	30	32	36	37	37	36	35	30	23	19	19
All-Sky Solar Radiation	RadAvg		495	854	1210	1516	1693	1840	1963	1652	1228	750	455	346
	RadStd		54	82	120	136	117	128	87	132	92	112	48	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-4.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page